

数学作业纸

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10.6.1 (a) $\varphi_f = 0^\circ$ 阻抗, $\varphi_a \rightarrow - \varphi_a = 180^\circ$.

$\varphi_a + \varphi_f \neq 360^\circ$ 不满足

(b) 阻抗, $\varphi_a = 360^\circ$

$\varphi_a + \varphi_f = 360^\circ$ 满足.

10.6.2 (1) 阻抗, $\varphi_a \rightarrow - \varphi_a$, 满足.

(2) $|A| > 3$.

$$V_f = \frac{R_{e1}}{R_{e1} + R_f} V_o = V_i$$

$$|A| = \frac{V_o}{V_i} = 1 + \frac{R_f}{R_{e1}} > 3 \Rightarrow R_f > 2R_{e1}. f_0 = \frac{1}{2\pi RC} \approx 49.8 \text{ Hz}.$$

R_f 可为电阻, 也可以是电感.

10.6.5 (1) $\varphi_a + \varphi_f = 360^\circ$ (2) $|A| > 3$. $V_f = \frac{R_1}{R_1 + R_f + R_2} V_o = V_i$.

$$|A| = \frac{V_o}{V_i} = 1 + \frac{R_f + R_2}{R_1} \Rightarrow R_f + R_2 > 2R_1.$$

$$(3) f_0 = \frac{1}{2\pi RC} \approx 1591.5 \text{ Hz}.$$

$$(4) V_{om} = V_{RP} + V_N$$

$$V_N = V_P = V_i = \frac{1}{3} V_{om}$$

$$I_{R1} = I_{RP} = \frac{V_N - 0}{R_1} = \frac{\frac{1}{3} V_{om}}{R_1}$$

$$V_{RP} = \frac{\frac{1}{3} V_{om}}{R_1} R_P$$

$$V_{om} = \frac{1}{2} \frac{V_{om}}{R_1} R_P + \frac{1}{3} V_{om}$$

$$V_{om} = \frac{3R_1}{2R_1 - R_P} V_N$$

$$10.6.6. A = \frac{R_1(R_1 + R_2 + R_P)}{R_1} \approx 3. P_P = 3.55 \text{ kW}.$$

$$10.8.3 \text{ (a)} \cdot \frac{V_I - V_N}{R_1} + \frac{V_{REF} - V_N}{R_2} = 0. V_N = \frac{R_1}{R_1 + R_2} V_{REF} + \frac{R_2}{R_1 + R_2} V_I = 0.$$

$$V_I = V_{I2} = -\frac{R_1}{R_2} V_{REF} \approx 1 \text{ V}$$

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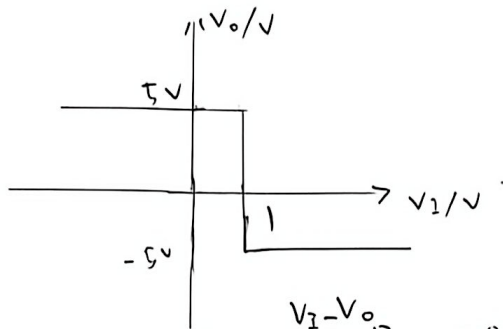
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$$10.8.4 \quad V_p = V_1 - V_{R1} = V_I - \frac{V_I - V_0}{R_1 + R_2} R_1 = 0$$

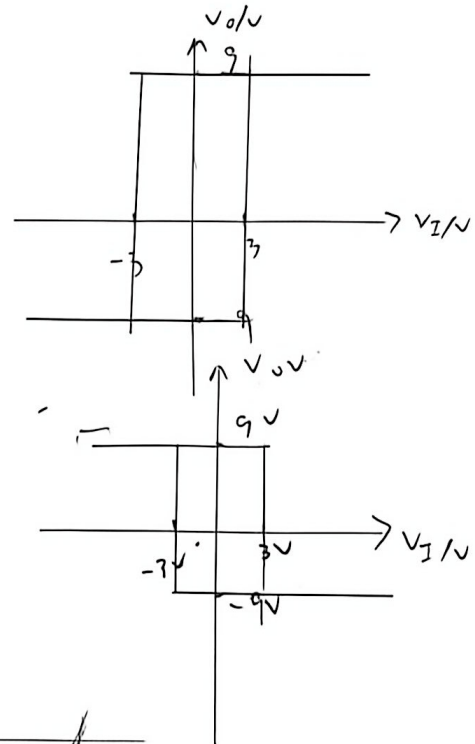
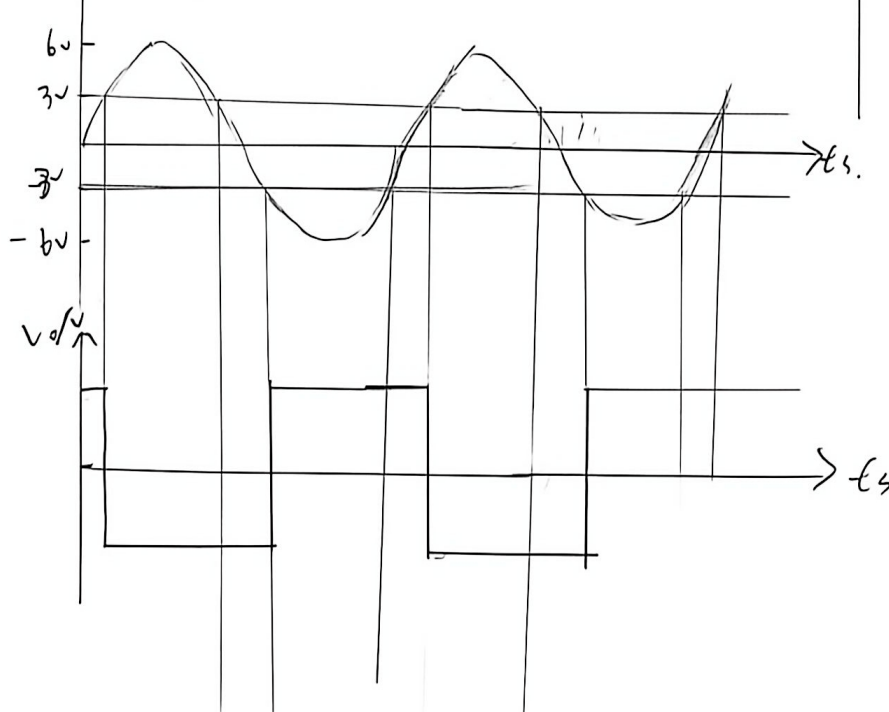
$$V_I = -V_0 \frac{R_1}{R_2} = \pm 3V$$

$$10.8.6.111 \quad V_0 = V_Z + V_p$$

$$V_p = V_{R1} = \frac{R_2}{R_1 + R_2} V_0 = \frac{V_0}{3}$$

$$V_0 = V_Z + \frac{V_0}{3} \Rightarrow \frac{2}{3} V_0 = 9V \Rightarrow V_0 = 13.5V$$

$$V_I/V \quad V_I = \frac{1}{3} V_0 = \pm 4.5V$$



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10.8.7. (1) $V_A = V_{CC} - \frac{2V_{CC}}{2R_2 + R_3} R_2 = \frac{R_3}{2R_2 + R_3} V_{CC}$. 设 $R_2 = R_1 = 0.1k\Omega$ $R_1 = 1k\Omega$ $R_2 = 200k\Omega$
 $V_{CC} = 12V$
 $\approx 4V$.

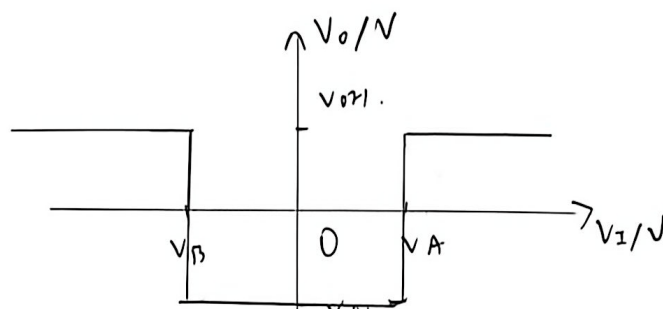
$V_B = V_{CC} - \frac{2V_{CC}}{2R_2 + R_3} (R_2 + R_3) = -4V$.

$V_I > V_A$ 时, D_2 通, V_I 进入同相端, $V_N = V_A$, 输出 V_{OH} .

$V_I < V_B$ 时, D_1 通, $V_I = V_N$, $V_P = V_B$, 输出 V_{OH} .

$V_B < V_I < V_A$ 时, D_2, D_1 通, 输出 V_{OL} .

(2).

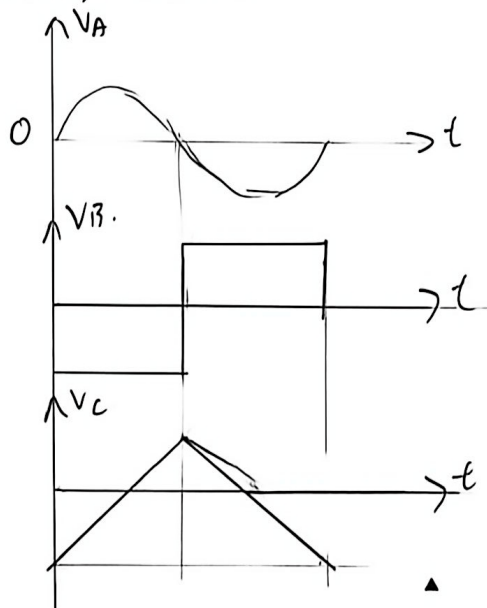


10.8.8. A_1 为 RC 移相式正弦波振荡器 $\rightarrow$ 输出正弦

A_2 为过零比较器 $\rightarrow$ 输出方波.

A_3 为鉴相器 $\rightarrow$ 输出三角波.

A_4 与 A_5, A_6 为电压跟随器 $\rightarrow$ 提升带负载能力.



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$$10.8.9 \quad V_{P1} = \frac{R_2}{R_1 + R_2} V_{in} + \frac{R_1}{R_2 + R_1} V_{O1} \quad \frac{1}{2} V_{P1} = V_{in} \quad \checkmark$$

$$V_T = V_{in} = \frac{R_1}{R_2} V_{O1} \quad V_{O1} = \pm V_Z$$

$$V_{T+} = 2.72V \quad V_{T-} = -2.72V$$

$$V_{O1} = V_Z \text{ 时, } V_{O2} = -\frac{V_Z}{R_1 C} (t_1 - t_0) + V_{O2}(t_0)$$

$$V_{O1} = -V_Z \text{ 时, } V_{O2} = \frac{V_Z}{R_1 C} (t_1 - t_0) + V_{O2}(t_0)$$

$$V_{opp} = 5.14V$$

$$\frac{1}{C} \int_0^T \frac{V_Z}{R_1} dt = V_{opp}$$

$$T = 2R_1 C \frac{V_{opp}}{V_Z} = \frac{4R_1 R_2}{R_2}$$

$$f = \frac{1}{T} = 3067.6 \text{ Hz}$$

$$10.8.10 \quad V_{O1} = -\frac{R_3}{R_1} V_1 - \frac{R_3}{R_2} V_{PEF} = -0.15V$$

$t = 1.4 \mu s$

$$V_{O2} = -\frac{1}{R_5 C} \int_0^t V_{O1} dt = 1.5V$$

$$V_T = \pm \frac{R_7}{R_8} V_Z = \pm 1.2V$$

$$t = 1.4 \mu s, V_{O2} > V_{T+}, V_{O3} = 6V$$

